Comparison: Aptara MCS SSRS Standard Product

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Contents

Note

Changes from Revision 000 to Revision 001.
~~Red strikethrough text was deleted.~~
Green text was added.

1 Document Overview

This document defines the standard product requirements for the Aptara Modular Control System (MCS) Subsea Supervisory and Reporting System (SSRS). It provides a common technical baseline for system design, implementation, verification, and project-specific configuration.

1.1 Purpose

The purpose of the SSRS is to collect operational data from connected control system components, present system status to authorized users, and retain the information required for engineering analysis and operational reporting.

> **Note**
>
> ~~Revision 000 is the initial controlled baseline. Project-specific deviations shall be documented separately and shall not change the standard product requirements without formal review.~~
> Revision 001 adds availability monitoring, backup integrity, and recovery verification requirements. Project-specific deviations shall be documented separately and shall not change the standard product requirements without formal review.

1.2 Scope

The standard SSRS product includes:

- Acquisition of status, alarm, event, and process data.
- Operator access to current and historical system information.
- Generation of standard operational and maintenance reports.
- Controlled exchange of data with approved external systems.
- User authentication and role-based authorization.

The following items are outside the scope of this document:

- Project-specific tag databases.
- Customer network infrastructure outside the SSRS boundary.
- Third-party reporting packages not supplied as part of the standard product.

2 System Description

2.1 Functional Architecture

The SSRS consists of application services, data services, operator interfaces, and communication interfaces. The system shall support deployment on an approved industrial computing platform and shall operate without requiring continuous access to an external cloud service.

Table 2-1: Standard SSRS functional components

Component	Primary Function	Typical Interface
Data Acquisition Service	Collect and validate source data	OPC UA or approved project protocol
Historian	Retain time-series values and events	Internal data service
Reporting Service	Generate scheduled and on-demand reports	Web user interface
Operator Interface	Display status, alarms, and trends	HTTPS
Administration Service	Manage users, roles, and configuration	Restricted HTTPS interface

2.2 Operating Modes

The SSRS shall support the following operating modes:

1. **Normal operation** - all configured services are available and data is collected continuously. ~~2. **Degraded operation** - one or more external interfaces are unavailable,~~

~~while local services continue operating where possible.~~

2. **Degraded operation** - one or more external interfaces are unavailable.

Local data collection and reporting services shall continue operating where

their required dependencies remain available.

2. **Maintenance mode** - authorized personnel may stop selected services for backup, configuration, or software maintenance.

3. **Recovery mode** - services and retained data are restored following a controlled recovery procedure.

3 Product Requirements

3.1 General Requirements

ID	Requirement
SSRS-GEN-001	The SSRS shall display the current availability of each configured data source.
SSRS-GEN-002	The SSRS shall time-stamp collected records using the configured system time source.
SSRS-GEN-003	The SSRS shall retain an audit record of security-relevant administrative actions.
SSRS-GEN-004	The SSRS shall recover automatically after restoration of the computing platform.
SSRS-GEN-005	The SSRS shall prevent unauthorized modification of controlled configuration data.
SSRS-GEN-006	The SSRS shall monitor critical application services and report an unavailable service to authorized users.
SSRS-GEN-007	The SSRS shall verify the integrity of a completed configuration backup before reporting the backup as successful.

3.2 Data and Reporting Requirements

ID	Requirement
SSRS-DAT-001	The SSRS shall retain process values at the configured sampling interval.
SSRS-DAT-002	The SSRS shall preserve alarm and event time order during normal operation.
SSRS-DAT-003	Authorized users shall be able to export approved report data in PDF and CSV formats.
SSRS-DAT-004	Reports shall identify the report period, generation time, and data source.
SSRS-DAT-005	Buffered records shall be forwarded in chronological order after restoration of an interrupted data-source connection.

3.3 Security Requirements

ID	Requirement
SSRS-SEC-001	Each interactive user shall authenticate using an individually assigned account.
SSRS-SEC-002	Access permissions shall be assigned through approved user roles.
SSRS-SEC-003	Communication with browser-based clients shall use encrypted HTTPS sessions.
SSRS-SEC-004	Security logs shall be available to authorized administrators.

4 External Interfaces

4.1 Control System Interface

The control system interface shall provide the configured process values, equipment states, alarms, and events. Loss of this interface shall be reported ~~to the operator and shall not corrupt previously retained data.~~

to the operator within the configured monitoring interval and shall not corrupt

previously retained data.

4.2 User Interface

The user interface shall provide navigation to dashboards, trends, alarms, reports, and administrative functions according to the signed-in user's role.

4.3 Time Synchronization

The SSRS computing platform shall synchronize with the approved project time source. All displayed and exported timestamps shall identify the configured time basis.

5 Verification

~~The revision 000 baseline shall be verified using the methods below.~~

The revision 001 baseline shall be verified using the methods below.

Table 5-1: Initial verification matrix

Verification Area	Method	Acceptance Criterion
Installation	Inspection	Required services and configuration files are present.
Data acquisition	Test	Configured values are received and time-stamped correctly.
Alarms and events	Test	Generated test events appear in the correct sequence.
Reporting	Demonstration	An authorized user can generate and export a standard report.
Access control	Test	Each test role can access only its permitted functions.
Recovery	Test	Services recover after a controlled platform restart.
Backup integrity	Test	A valid backup is accepted and an intentionally corrupted backup is rejected.
Interface recovery	Test	Buffered records are transferred in time order after communication is restored.

6 Revision Notes

~~Revision 000 establishes the initial SSRS Standard Product baseline. Detailed project configuration, final performance limits, and customer-specific interfaces will be defined in the applicable project documentation.~~

Revision 001 extends the SSRS Standard Product baseline with service-availability monitoring, backup-integrity verification, and deterministic recovery of buffered records after an interface interruption.